

KEEPSAKE

Updated Vision & Design Bible

v3.0 - Vision, Product Systems, Technical Direction, and Current Build State

Purpose	A readable source of truth for the product: what it is, how it should feel, what belongs, what does not, how each major system should behave, and how to build it in a sensible order.
Current build	A working React + TypeScript + Vite scrapbook prototype exists. The book opens/closes, spreads navigate, photos upload to left/right pages, and photos can be selected, dragged, rotated, resized, reset, replaced, and removed.
Build philosophy	Prove the scrapbook experience first. Add the room, accounts, social systems, progression, music, offline support, printing, and long-term systems only after the solo memory loop is stable and satisfying.
Notation	Yellow OPEN DECISION callouts identify unresolved design or engineering choices that should be settled before production implementation.

1. Executive Summary

Core promise: A calm, tactile digital scrapbook that turns a user's photographs into a place they can revisit, not a feed they must keep up with.

- The room is the interface. Major systems are represented by physical objects when practical.
- Photographs are always the hero. Decoration, music, achievements, social interaction, and ambience exist to support memories.
- Every user begins with the same starter room; over time the room molds to the person through customization, earned objects, scrapbook styles, memory patterns, and environmental changes.
- The product begins as a solo scrapbook experience and may grow into a restrained social platform where friends visit authorized rooms, browse books, sign a guest book, leave notes, and create subtle room traces.
- The design intentionally rejects follower counts, infinite feeds, daily quests, virtual currency, pay-to-win progression, intrusive notifications, and NPC-driven distraction.

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2. North Star and Laws of the House

North star	The product should feel like entering a personal memory space, opening a real scrapbook, and spending time with photographs rather than consuming content.
Priority order	1) Photographs, 2) Scrapbook, 3) Organization, 4) Room atmosphere, 5) Safe visiting/social, 6) Progression/cosmetics.
Emotional tone	Cozy, nostalgic, handcrafted, calm, slightly magical, but never noisy, childish, manipulative, or overloaded.
Interaction test	If an interaction can be expressed as a room object or physical action without creating friction, prefer the physical metaphor.

- Every major feature should feel like a physical object or action in the room.
- Every interaction should support storytelling, memory, organization, atmosphere, or safe visiting.
- Discovery is more rewarding than instruction.
- The House Has Memory: the room can preserve context such as an open book, recently viewed memory, saved notes, and long-term personalization.
- The House Has Manners: the room does not interrupt active reading, editing, emotional moments, or other important actions.
- The House Pays Attention: ambient events are selected based on context rather than purely random timing.
- Curated RNG beats raw randomness. Eligible events are filtered by context, cooldowns, user preferences, and quiet-state rules.
- One beautiful thing is better than five simultaneous events. Silence is valid.
- Never make the user wait for immersion. Let immersion happen while they are already doing something meaningful.
- The room should gradually become a portrait of its owner without requiring constant manual customization.

Design test: Before adding a feature, ask whether it makes memories easier to preserve, organize, revisit, personalize, or share thoughtfully. If not, it probably does not belong.

3. Current Build State and Code Organization

Frontend	React + TypeScript + Vite.
Working behavior	Closed cover; open two-page spread; previous/next navigation; keyboard navigation; file upload; independent left/right page photos; photo selection; drag; scale; rotation; reset; remove; controlled protrusion while open; no photo protrusion while closed.
Persistence	Temporary in-memory state. Uploaded images disappear on refresh. This is intentional at the current prototype stage.
Styling	CSS-based pseudo-3D scrapbook with perspective, shadows, paper textures, photo frames, and limited edge overlap.
Refactor	The codebase is being split into components, data, types, utilities, and later hooks before multi-photo page complexity is added.

Recommended Near-Term File Structure

```
src/  
  components/  
    BookCover.tsx  
    ScrapbookPage.tsx  
    PhotoItem.tsx  
    PhotoToolbar.tsx  
  data/  
    scrapbookData.ts  
  types/  
    scrapbook.ts  
  utils/  
    clamp.ts  
  hooks/  
    useScrapbookPhotos.ts  
  App.tsx  
  App.css  
  index.css  
  main.tsx
```

Why refactor now: The prototype already mixes uploads, transforms, selection, navigation, keyboard controls, and rendering. Separating responsibilities before multi-photo pages prevents App.tsx from becoming a 1,500-line file.

4. Technical Architecture

Frontend	React + TypeScript + Vite. Add React Router when user accounts and separate routes are introduced.
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3D layer	React Three Fiber + Three.js. Blender for authored room, desk, books, props, page meshes, and environment assets.
State	Use React state/useReducer during prototype. Add Redux Toolkit only when room/editor/offline/event state becomes hard to coordinate.
Backend	Node.js + TypeScript with Fastify preferred; Express is acceptable if the coding partner is more comfortable with it.
Database	PostgreSQL. Prisma or Drizzle ORM.
Photo storage	Cloudflare R2 or another S3-compatible object store. Store originals privately and generate smaller derivatives.
Auth	Better Auth or Auth.js for lower lock-in; Clerk remains a valid managed alternative.
Hosting	Render or Railway for production services; Vercel may remain a preview/testing environment.
Testing	Vitest + React Testing Library + Playwright.
Art/design	Figma for interface planning; Blender for 3D; licensed asset libraries where they save production time.

OPEN DECISION / NEEDS MORE ATTENTION: Finalize Fastify vs Express and Prisma vs Drizzle with the coding partner before backend implementation. Consistency matters more than the exact choice.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether the first room is full 3D or a 2.5D scene with authored camera views. 2.5D is faster, simpler, and safer for mobile.

5. Core Scrapbook Experience

Purpose	Deliver the primary memory experience. If the scrapbook is not enjoyable by itself, the room and social systems do not matter.
Experience	Cover -> open -> view spreads -> navigate -> add photos -> arrange them -> optionally caption/decorate -> save -> revisit later. The title page introduces the book before the photographic story begins.
Implementation	Represent scrapbooks, spreads, pages, and page elements as data. Keep viewer rendering efficient and editor rendering interactive. Provide presets for fast users and freeform placement for users who enjoy craft.
Software / stack	React, TypeScript, CSS initially; later Konva.js or Fabric.js may be useful for a richer page editor.
Core data	scrapbooks, spreads, pages, page_elements, page_order, title_page, caption, visibility.
Approx. timeline	Current prototype already exists. Functional multi-photo MVP: 3-6 additional weeks. Polished editor: another 4-8 weeks.
Main risks	Editor complexity, undo/redo, mobile gestures, page element collisions, preserving layout across screen sizes.
Definition of done	A user can create a multi-page book, place 2-6 photos per page, navigate reliably, save, reload, and see the same layout.

OPEN DECISION / NEEDS MORE ATTENTION: Choose the long-term page editor approach: custom DOM transforms vs Konva.js vs Fabric.js.

6. Photo Upload, Editing, and Multi-Photo Pages

Purpose	Reliably ingest photographs and let users arrange them in a tactile way.
Experience	Photos feel like physical prints. They may overlap, rotate, resize, and slightly protrude from an open page, but nothing appears outside the closed book.
Implementation	Prototype: FileReader + React state. Production: signed upload to object storage, metadata row, EXIF orientation normalization, derivatives, private originals. Page photos become arrays of stable photo objects with unique IDs, transforms, crop metadata, and z-order.
Software / stack	Browser File API; Node image processing (Sharp); Cloudflare R2; PostgreSQL; optional Konva/Fabric for editing.
Core data	photos, photo_files, derivatives, page_photo_elements, x, y, rotation, scale, crop, z_index, original_layout.
Approx. timeline	Multi-photo page: 1-2 weeks. Production upload pipeline: 2-4 weeks. Advanced crop/editor tools: later.
Main risks	HEIC support, large files, mobile memory use, duplicate uploads, object URL cleanup, complex overlapping photos.

Definition of done	Each page supports several independent photos; each can move/rotate/resize/replace/delete; production photos survive refresh and use optimized display files.
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OPEN DECISION / NEEDS MORE ATTENTION: Set final page-photo limit after usability testing. Start with six.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether captions have protected no-overlap zones.

7. Page Turning and Physicality

Purpose	Make navigation tactile without forcing the user through slow cinematics.
Experience	The book subtly lifts, the page bends/curls, a shadow moves, and the spread settles. Keyboard/touch users receive the same result without friction.
Implementation	Use a state machine: idle -> lifting -> turning -> settling. Preload adjacent spreads. Begin with directional CSS/DOM animation; only move to a 3D page mesh if the visual payoff justifies complexity.
Software / stack	CSS animations, React state machine; later React Three Fiber / Three.js / Blender page mesh.
Core data	current_spread, direction, animation_state, reduced_motion_preference.
Approx. timeline	Directional CSS pass: 2-5 days. Convincing 3D page turn: 2-5 weeks of iteration.
Main risks	Texture swapping, animation interruption, page content distortion, mobile performance.
Definition of done	Page turns are responsive, preloaded, directional, interrupt-safe, and have a reduced-motion fallback.

OPEN DECISION / NEEDS MORE ATTENTION: Do not commit to Three.js page turning until multi-photo page rendering is stable.

8. Room, Camera, and Clutter Control

Purpose	Create the spatial home around the scrapbook without allowing features to overwhelm the photographs.
Experience	Everyone starts with the same room. Users move between authored viewpoints: desk, scrapbook, shelf, archive, CRT, timeline, map, window. The room feels alive but quiet.
Implementation	Model a modular room in Blender and export optimized GLB assets. Use fixed camera targets and a CameraDirector rather than free-roaming controls. Create an object interaction registry and a strict clutter budget per view.
Software / stack	Blender, React Three Fiber, Three.js, glTF/GLB, KTX2/texture compression, optional GSAP.
Core data	room_preferences, object_visibility, room_theme, camera_preference, decor_slots, unlocks.
Approx. timeline	Prototype room: 2-4 weeks. Polished optimized room: 1-3 months depending on art production.
Main risks	3D asset creation, mobile GPU performance, lighting cost, visual clutter, camera motion sickness.
Definition of done	The room loads within budget, every core object is reachable, the scrapbook remains visually dominant, and low-power/reduced-motion modes work.

OPEN DECISION / NEEDS MORE ATTENTION: Define a formal visual object budget for each camera view.

OPEN DECISION / NEEDS MORE ATTENTION: Choose full 3D vs 2.5D for MVP.

9. Bookshelf and Multiple Scrapbooks

Purpose	Make multiple albums feel like a growing physical library rather than a dropdown.
Experience	Users select books by spine/cover. Books may represent years, trips, people, or themes. The current book closes before another is brought to the desk.
Implementation	Reuse a small set of optimized book meshes and generate spine title textures/material variants. Load only shelf metadata until a book is opened.
Software / stack	React Three Fiber, Blender, Canvas texture generation, backend metadata API.

Core data	scrapbooks, shelf_order, spine_style, cover_style, visibility, archived_state.
Approx. timeline	1-2 weeks after the core scrapbook system is stable.
Main risks	Shelf overflow, unauthorized book titles, visual readability of spine text.
Definition of done	Owner can create/reorder/archive books; visitors see only permitted books; shelf remains usable with many albums.

OPEN DECISION / NEEDS MORE ATTENTION: Define how shelf overflow is represented: additional shelves, boxes, or navigable shelf sections.

10. Filing Cabinet and Photo Archive

Purpose	Organize unused and older photographs without turning the room into a file manager.
Experience	The most recent 10-20 uploads remain as a physical stack in a drawer. Older photos live in a filing cabinet under user-defined categories and smart folders.
Implementation	Use normal indexed database search and cursor pagination behind a physical folder/tab UI. Dragging a photo into a scrapbook creates a reference to the existing photo asset rather than duplicating it.
Software / stack	PostgreSQL full-text search, React, optional virtualized list, React Three Fiber for the cabinet interaction.
Core data	photo_collections, collection_members, tags, recent_upload_order, photo_usage, favorites.
Approx. timeline	Recent stack: 3-5 days. Filing/search system: 2-4 weeks.
Main risks	Large libraries, search speed, tagging burden, metadata privacy.
Definition of done	A 10,000-photo metadata library remains searchable and usable; recent uploads are obvious; archives are category-driven and fast.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether people tags are manual only at launch.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether location metadata is imported automatically or opt-in.

11. Craft Drawer, Stickers, and Scrapbook Styles

Purpose	Add creative expression without compromising the central role of photographs.
Experience	Users open a craft drawer or sticker book rather than a generic asset shop. Starter packs are generous; seasonal drops are free and recurring; paid packs add style only.
Implementation	Treat stickers as transformable page elements sharing x/y/rotation/scale/z-index behavior with photos. Store pack entitlements separately from progression. Use lightweight SVG/WebP assets.
Software / stack	Figma/Illustrator, asset pipeline, React page editor, Stripe later for commerce.
Core data	asset_packs, assets, entitlements, seasonal_availability, page_elements.
Approx. timeline	Starter system: 2-3 weeks. Commerce: 2-4 additional weeks.
Main risks	Page clutter, licensing, asset size, accidental pay-to-win signals.
Definition of done	Users receive 3-5 starter sticker packs and several scrapbook styles; packs can be earned or purchased; paid packs never affect progression.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether user-uploaded stickers will ever be allowed.

12. Hidden Achievements and Progression

Purpose	Reward care, craft, organization, and memory preservation without turning the product into a checklist.
Experience	Most achievements are Easter eggs. Users discover rewards through letters, new supplies, object changes, CRT messages, found keepsakes, or other environmental presentations.
Implementation	Separate achievement completion from reward presentation. Evaluate hidden rules from user-owned actions, record immutable completion, grant rewards idempotently, and allow curated RNG to choose how/when the reward appears.
Software / stack	PostgreSQL rules, Node/Fastify jobs, event ledger, entitlement service.

Core data	achievement_definitions, achievement_progress, completion, reward_grants, delivery_receipts.
Approx. timeline	Framework: 2 weeks. Initial 20-30 achievements/rewards: 2-4 weeks plus writing/art.
Main risks	Players reverse-engineering rules, repetition, reward spam, accidental incentives to grind.
Definition of done	Achievements never depend on spending, likes, follower counts, or visitor volume; rewards are reliably granted; presentation remains varied and optional.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether progression titles are public, private, or opt-in.

13. CRT Music, Spotify, and Offline Audio

Purpose	Make music part of the memory atmosphere while preserving a complete experience without external accounts.
Experience	The CRT is the music interface. Users can link Spotify online, associate playlists with scrapbooks, or choose licensed lo-fi/ambient tracks for offline mode.
Implementation	Separate the CRT visual controller from playback providers. Use provider adapters: Spotify online and local licensed-track playback offline. Persist volume, mute, playlist reference, and per-book selection.
Software / stack	Spotify Web API/SDK, Howler.js or Web Audio API, licensed royalty-free music files, service worker caching.
Core data	music_preferences, playlist_reference, offline_track_id, per_scrapbook_music, volume.
Approx. timeline	Offline player: 1 week. Spotify linking/playback: 2-4 weeks plus policy review.
Main risks	Music licensing, autoplay restrictions, Spotify policy changes, offline rights.
Definition of done	Music never blocks the scrapbook, respects user settings, has a clear mute, and offline tracks work without network.

OPEN DECISION / NEEDS MORE ATTENTION: Review Spotify platform terms and music licensing before public launch.

OPEN DECISION / NEEDS MORE ATTENTION: Do not support user-uploaded copyrighted music at launch.

14. Window, Day/Night, Seasons, and Ambience

Purpose	Make the room feel alive through subtle environmental change.
Experience	Users can choose day/night and season, or optionally follow time. The window, room lighting, lamp, ambience, and small environmental details respond.
Implementation	Use a small authored set of lighting/environment presets rather than real-time simulation. Swap layered window/environment textures and particle systems. Separate ambience, music, and interaction audio channels.
Software / stack	Blender, React Three Fiber, Three.js lights/materials, Web Audio/Howler.
Core data	environment_preference, season, time_mode, weather_mode, ambience_volume.
Approx. timeline	Initial day/night: 3-5 days. Four seasons + ambience: 2-4 weeks.
Main risks	Too many particles/lights, distracting animation, mobile performance.
Definition of done	Season/time changes are visually clear but never interfere with reading; reduced-motion and low-power modes simplify effects.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether live weather is ever needed. Manual/curated weather is safer and simpler.

15. Timeline Wall

Purpose	Show the user's life as a connected sequence rather than only separate albums.
Experience	The camera pans to a wall where years and selected milestones appear. Selecting a year can highlight related scrapbooks and archive memories.
Implementation	Aggregate dates server-side. Render decades/years first, then reveal milestones on focus. Keep the wall sparse and user-curated or lightly

	suggested.
Software / stack	React Three Fiber or SVG/canvas texture, PostgreSQL date aggregation.
Core data	timeline_entries, year, milestone_type, linked_scrapbook, pinned_memory.
Approx. timeline	Basic yearly timeline: 1-2 weeks. Rich interactions: 2 more weeks.
Main risks	Long lifespans, missing dates, clutter, automatic suggestions feeling intrusive.
Definition of done	The timeline stays readable from a few years to decades of content and respects all privacy rules.

OPEN DECISION / NEEDS MORE ATTENTION: Define whether timeline milestones are user-only, suggested, or mixed.

16. Memory Atlas / World Map

Purpose	Let users associate one representative memory with meaningful places.
Experience	A separate wall contains an illustrated map/corkboard. Users add pins, one photo, a short caption, and optional date/location label.
Implementation	Store a location record with coordinates or approximate location. Render through an illustrated map layer. Keep exact coordinates private by default and allow manual placement/approximation.
Software / stack	React Three Fiber or SVG, mapping/geocoding service only if needed, PostgreSQL.
Core data	memory_pins, label, approximate_coordinates, photo_id, caption, visibility.
Approx. timeline	Manual illustrated map: 2-3 weeks. Search/geocoding: 1-2 extra weeks.
Main risks	Location privacy, geocoding costs, map licensing, exact EXIF leakage.
Definition of done	Pins are easy to add/remove, privacy is independent, and visitors never receive exact location data unless explicitly authorized.

OPEN DECISION / NEEDS MORE ATTENTION: Use approximate/manual locations by default; exact coordinates should be opt-in.

17. Accounts, Ownership, Privacy, and Friend Levels

Purpose	Make owner vs visitor behavior secure, understandable, and flexible.
Experience	Users understand who can enter, view, listen, write, or leave traces. Defaults are conservative; advanced controls exist without overwhelming setup.
Implementation	Implement permission inheritance: account default -> room -> scrapbook -> page -> item/action. Enforce permissions server-side/database-side, not only by hiding buttons. Add View As Public/Friend/Close Friend preview.
Software / stack	Better Auth/Auth.js, Fastify, PostgreSQL, type-safe permission engine.
Core data	users, profiles, relationships, permission_defaults, overrides, blocks, invites.
Approx. timeline	Core auth: 1-2 weeks. Permission engine/UI/tests: 3-5 weeks.
Main risks	Privacy leaks, complex inheritance bugs, storage URLs bypassing rules.
Definition of done	Unauthorized media cannot be fetched; every action resolves to a deterministic permission result; owner can preview each tier.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether public rooms are searchable or direct-link-only at initial launch.

18. Visitor Modes and Room Presence

Purpose	Make social visits feel like entering someone's home instead of browsing a profile.
Experience	Visitors may leave no trace, restore the room, leave the last authorized book/page open, or leave a permitted note. Owners choose which modes are allowed.
Implementation	Create lightweight visitor sessions with entry, mode, viewed authorized objects, and optional exit trace. Never store or expose live cursor trails. Validate every trace against owner permissions.
Software / stack	React, Fastify, PostgreSQL; optional realtime later.
Core data	visitor_sessions, visitor_mode, exit_trace,

	last_viewed_authorized_object, retention_expiry.
Approx. timeline	Basic visits: 1 week. Trace behaviors/preferences: 1-2 more weeks.
Main risks	Privacy, high traffic, stale traces, visitors modifying persistent content.
Definition of done	Visitor traces are reversible, permission-safe, optional, and do not change permanent owner content.

OPEN DECISION / NEEDS MORE ATTENTION: Define visitor session retention and whether close friends can visit while the owner edits.

19. Guest Book, Notes, Whispers, and High-Traffic Rooms

Purpose	Create meaningful social connection without comments, threads, counters, or notification pressure.
Experience	General visitors sign a guest book. Trusted visitors may leave page-linked handwritten notes. Approved notes may occasionally appear as whispers or catch on the windowsill.
Implementation	Guest book entries are flat/non-threaded. Page notes are limited physically to prevent clutter. Whispers select only approved entries, respect frequency and owner threshold, avoid repeats, and never interrupt active interactions.
Software / stack	PostgreSQL, moderation/rate-limit layer, React Three Fiber for ambient presentation.
Core data	guestbook_entries, notes, target_page, approval_state, whisper_eligibility, shown_history, threshold.
Approx. timeline	Guest book: 2 weeks. Notes/whispers: 2-3 more weeks.
Main risks	Spam, celebrity-scale volume, clutter, moderation burden, quote repetition.
Definition of done	High-traffic rooms remain quiet; all messages are reviewable in the guest book; only a few curated notes appear physically.

OPEN DECISION / NEEDS MORE ATTENTION: Define the visitor threshold using unique visitors rather than total visits.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether surfaced quotes display username, initials, or anonymous attribution.

20. Curated RNG / House Behavior

Purpose	Create rare, context-aware surprises that make the room feel attentive without becoming a game loop.
Experience	The room may leave an old scrapbook slightly open, surface a meaningful note, present an achievement reward, or quietly change a small object. Often nothing happens.
Implementation	Use an opportunity queue. Each candidate has eligibility, priority, cooldown, emotional intensity, quiet-state requirements, delivery methods, and privacy checks. A curator selects zero or one event and records a receipt.
Software / stack	TypeScript rules engine, Fastify scheduled jobs, PostgreSQL event ledger, local presentation queue.
Core data	event_candidates, event_receipts, cooldowns, house_preferences, delivery_method.
Approx. timeline	Rules engine: 1-2 weeks. First event catalog: 2 weeks. Ongoing tuning thereafter.
Main risks	Repetition, unpredictability feeling manipulative, event collisions, debugging randomness.
Definition of done	At most one ambient surprise appears in a short session window; events honor cooldowns/preferences; tests can reproduce choices with seeded randomness.

OPEN DECISION / NEEDS MORE ATTENTION: Define the first 10-20 ambient opportunities before writing the engine.

21. Environmental Notifications and Storytelling

Rule	Notifications should not interrupt the room. Whenever practical, information appears through environmental change or a physical object.
Examples	Letter on desk; sticky note inside book; note caught on windowsill; new sticker book in drawer; book left slightly open; seasonal gift object; CRT message.

Typography	Inspired by environmental storytelling: short words/choices may float or appear in the scene instead of an ugly modal dialog.
Fallback	Important account/security/moderation notices still need a conventional accessible interface outside the immersive room.

- Floating text should be short, optional, dismissible by ignoring it, and never block input.
- Do not display a whisper during page turns, photo editing, milestone reading, or another environmental event.
- The user controls whisper frequency and may disable environmental notifications entirely.
- Environmental messages should have a conventional accessible equivalent in a guest book/settings/activity area when retention matters.

22. Performance, Scalability, and Offline Mode

Purpose	Keep the experience fast even with large photo libraries and visually rich rooms.
Experience	The user sees meaningful content first. Decorative immersion loads around it. Selected books can be saved for offline read-only access with licensed music.
Implementation	Set budgets for JS, GLB, textures, image derivatives, frame time, and API latency. Lazy-load room zones and books. Precompute page textures for viewers where useful. Use service workers/IndexedDB for selected offline books.
Software / stack	Lighthouse, WebPageTest, Sentry Performance, React Profiler, Three.js tools, Workbox/service worker, IndexedDB.
Core data	performance_metrics, asset_versions, offline_manifest, cached_book_version, sync_state.
Approx. timeline	Instrumentation: 1 week. Offline read mode: 2-3 weeks. Ongoing optimization throughout project.
Main risks	Large images, low-power mobile devices, cache storage limits, 3D texture memory, stale offline permissions.
Definition of done	Target devices meet load/frame budgets; 10,000-photo account remains usable; selected books open offline; private cached data clears on logout.

OPEN DECISION / NEEDS MORE ATTENTION: Define target device classes and concrete performance budgets before the room grows.

23. Accessibility and Mobile

Purpose	Make every core task possible without precise pointer input, 3D motion, or sound.
Experience	Desktop feels like an open book. Mobile may show one page at a time with swipe/touch controls. Reduced-motion users get cuts/short fades instead of camera travel.
Implementation	Maintain semantic DOM equivalents for room objects; keyboard navigation; screen-reader labels; visible focus; non-drag alternatives for positioning; 200% zoom testing; touch gestures with adequate target sizes.
Software / stack	React ARIA patterns, axe-core, Playwright a11y tests, VoiceOver/NVDA manual testing.
Core data	accessibility_preferences, reduced_motion, text_scale, contrast, input_mode.
Approx. timeline	Foundation integrated continuously; dedicated audit 1-2 weeks before launch.
Main risks	3D-only controls, drag-only editing, small text, motion sickness, mobile layout mismatch.
Definition of done	All MVP tasks work with keyboard and without sound/animation; mobile remains usable; screen-reader path is complete.

OPEN DECISION / NEEDS MORE ATTENTION: Decide whether screen-reader mode skips the 3D room entirely and uses an object list.

24. Trust, Safety, Moderation, and Preservation

Purpose	Protect users, private memories, and long-term data integrity.
Experience	Owners can hide/delete/report/block social content. Backups and exports reduce fear of losing years of memories.

Implementation	Add rate limits, report categories, block enforcement, moderation states, upload scanning, audit logs, soft deletion, page revisions, scheduled backups, and export jobs.
Software / stack	Fastify, PostgreSQL, object storage versioning, moderation provider, malware scanning, admin interface.
Core data	reports, blocks, moderation_cases, audit_log, revisions, backup_jobs, export_jobs.
Approx. timeline	Basic block/report: 2 weeks. Public moderation tooling: 3-6 more weeks. Backup/export: 2-3 weeks.
Main risks	Spam, abusive notes, stolen media, privacy incidents, accidental deletion, moderation staffing.
Definition of done	Block applies everywhere; reported content can be removed from display; deleted pages can be restored; users can export their memories.

OPEN DECISION / NEEDS MORE ATTENTION: Define moderation staffing, appeals, retention, minimum age, and jurisdiction before public social launch.

25. Physical Scrapbook Printing and Export

Purpose	Let digital books become physical keepsakes and give users a portable archive.
Experience	Users can preview a print-ready version, see warnings for low-resolution photos, and eventually order a bound book.
Implementation	Generate a print-specific PDF from high-resolution originals using fixed trim sizes, bleed, safe zones, and image-resolution checks. Integrate a print-on-demand vendor only after proof testing.
Software / stack	Server-side PDF renderer, image processing, print vendor API, payment provider, background job queue.
Core data	print_jobs, export_jobs, print_validation, trim_size, page_render_version, shipping_status.
Approx. timeline	PDF/export renderer: 4-8 weeks. Vendor integration: 3-6 weeks plus contracting/testing.
Main risks	Color profiles, bleed, low-res images, shipping/refunds, vendor outages, expensive proof cycles.
Definition of done	Low-resolution pages are flagged before purchase; exported layouts match digital pages; vendor orders are traceable and recoverable.

OPEN DECISION / NEEDS MORE ATTENTION: Choose print vendor, supported countries, book sizes, binding options, and refund policy.

26. Monetization Rules

Core principle	Monetize expression and optional services, not memory preservation or progression.
Good candidates	Sticker packs, scrapbook themes, room cosmetic themes, optional decor packs, premium print products, possibly expanded storage if the free tier is still generous.
Never grant	XP, achievements, titles, social visibility, visitor reach, or progression from purchases.
Seasonal content	Free holiday/seasonal drops recur and should not use artificial scarcity or fear-of-missing-out pressure.

OPEN DECISION / NEEDS MORE ATTENTION: Earlier concepts included avoiding paid storage entirely. Decide whether storage will be a cost-recovery subscription or remain generous/free with premium cosmetics carrying more of the business model.

27. Explicit Do-Not-Build List

- No follower counts, public visit counts, popularity ranks, trending feeds, or engagement leaderboards.
- No infinite scrolling content feed.
- No daily quests, punitive streaks, loot boxes, virtual currency, or pay-to-progress.
- No NPCs in the planned product. Optional pets may be explored much later.
- No mini-games that distract from photographs.
- No autoplaying social alerts, red badges, or modal interruptions in the room.
- No unrestricted public comment threads or reply chains under photos.
- No giant in-house photo editor at launch. Start with crop/rotate/basic adjustments only if needed.

- No new permanent room object unless it supports a distinct memory, organization, atmosphere, or safe-visiting behavior and passes a clutter review.

28. Roadmap, Timelines, and Risk Ranking

Phase A - Current prototype	Working scrapbook cover, spreads, uploads, left/right photos, drag/rotate/scale/reset/remove. Refactor now.
Phase B - Scrapbook MVP	Multi-photo pages, page element model, presets, local persistence, better page turns, captions. Approx. 4-8 weeks.
Phase C - Solo persistent product	Backend, accounts, photo storage, saved books, bookshelf, archive, privacy. Approx. 2-4 months.
Phase D - Room alpha	3D/2.5D room, camera, window, CRT shell, shelf, cabinet, clutter/performance pass. Approx. 2-4 months depending on art.
Phase E - Social beta	Friends, permissions, visitor modes, guest book, notes, whispers, moderation. Approx. 2-4 months.
Phase F - Expansion	Achievements, craft marketplace, timeline, map, offline mode, Spotify, print ordering, legacy features. Ongoing.

Hardest / Highest-Risk Areas

1. Privacy and permission inheritance - errors can expose private media.
2. Production photo pipeline and large-library performance - images dominate storage, bandwidth, and browser memory.
3. Multi-photo freeform editor - transforms, undo/redo, mobile input, persistence, and layout consistency create many edge cases.
4. 3D room art/performance - functional code is easier than making it polished and fast across devices.
5. Trust/safety for public social use - spam, moderation, reporting, blocking, and operational workflows.
6. Offline editing/sync - read-only offline is manageable; conflict-safe offline editing is much harder.
7. Physical printing - partly a software problem and partly an operations/vendor problem.
8. Curated RNG tuning - technically straightforward to start, difficult to make consistently tasteful and nonrepetitive.

Realistic solo timeline: For a novice building part-time with AI assistance and a coding partner, expect roughly 12-24 months for a polished solo-user MVP and 2-4+ years for most of the full social vision. A focused small team can compress this significantly.

29. Immediate Next Coding Milestones

1. Finish the current refactor: BookCover, ScrapbookPage, types, data, clamp utility. Verify no behavior changes.
2. Extract photo rendering and drag logic into PhotoItem.tsx and a useScrapbookPhotos hook.
3. Change each page from one optional photo to an array of photos with unique IDs.
4. Add a page-level '+ Add photograph' control and support 2-6 photos per page.
5. Add z-order behavior, delete/replace selected photo, and sensible initial positions.
6. Add local persistence (localStorage or IndexedDB) for page layout only, while keeping image persistence temporary until production upload work begins.
7. Add editable short captions with character limit and positioning choices.
8. Improve page-turn direction and animation after the multi-photo data model is stable.
9. Only then begin the room shell; do not jump into 3D before the scrapbook editor proves enjoyable.

First Production-Ready Data Shapes

Scrapbook

- id
- ownerId
- title
- year/date range
- coverStyle
- visibility
- pageOrder[]

Page

- id
- scrapbookId
- index
- backgroundStyle
- elements[]

PageElement

- id
- type: photo | sticker | caption | keepsake
- x / y
- scale / width / height
- rotation
- zIndex
- element-specific payload

Photo

- id
- ownerId
- originalFile
- displayDerivative
- thumbnailDerivative
- metadata
- archiveCategories[]

Final Product Rule

Keep this true through every build phase: The scrapbook tells the story of the user's life. The room tells the story of the user's relationship with those memories. Technology should disappear behind both.

30. New Memory Systems - v4 Expansion

Expansion principle: These systems deepen existing objects rather than filling the room with new furniture. They should be designed into the data model now, but most should be implemented only after the core scrapbook editor is stable.

30.1 Memory Lighting

Purpose	Let the memory currently being viewed influence the room's atmosphere.
Experience	Opening a memory can subtly shift window light, desk-lamp warmth, ambience, and environmental tone. A beach memory might add warm sunset light and faint ocean ambience; a winter memory might cool the window and introduce soft wind.
Implementation direction	Store an optional memory mood profile containing season, time-of-day, setting, and tone. Begin with manual or user-approved tags. Later, optional image analysis may suggest a profile, but the user remains in control.
Core data	memory_mood_profile, season, time_of_day, environment_tag, intensity, enabled.
Guardrails	Do not make the effect literal or theatrical. It should feel subconscious. It must respect Lights Out, reduced-motion, sound preferences, and the House Has Manners rules.

30.2 Then & Now Pages

Purpose	Support deliberate visual comparisons between memories separated by time.
Experience	A user can pair childhood/adult photos, the same place years apart, family traditions, renovations, annual trips, or other repeated moments. The page may visually emphasize the dates and relationship.
Implementation direction	Treat Then & Now as a page layout/preset rather than a separate scrapbook type. Later, Keepsake may suggest possible pairs using date, location, tags, or user-confirmed subjects, but should never silently alter a book.
Core data	comparison_group_id, comparison_label, date_a, date_b, linked_photo_ids.
Guardrails	Suggestions must remain suggestions. Do not require facial recognition for this feature.

30.3 The Memory Box

Purpose	Give non-photographic memorabilia a first-class home inside Keepsake.
Experience	The Memory Box can contain concert tickets, movie/event tickets, newspapers and clippings, childhood artwork, drawings, handwritten notes, postcards, recipes, letters, programs, invitations, scanned documents, screenshots, and other meaningful objects.
Implementation direction	Create a Keepsake Artifact model separate from Photo. Users may upload scans/photos/PDFs, crop them, label them, preserve front/back sides, add date/place/context, and later place them onto scrapbook pages as physical-looking objects.
Core data	artifacts, artifact_files, artifact_type, front_asset, back_asset, title, date, place, notes, source, rights/provenance.
Guardrails	Artifacts should preserve the original uploaded source. Editing a scrapbook representation should never destroy the archival original.

30.4 AI Ticket Companion / Commemorative Stub

Purpose	Turn an Apple Wallet event pass or ticket information into a richer keepsake while preserving provenance and avoiding counterfeit-like representations.
Experience	A user can import or manually provide the event details visible in an Apple Wallet pass. Keepsake can search the web for historical references such as official event artwork, venue imagery, tour branding, or an authentic publicly available ticket image. If the user wants, Keepsake can instead create a clearly commemorative ticket-stub design inspired by the event.
Implementation direction	Use a provenance-first pipeline: extract user-authorized pass metadata; search trusted/public sources; present candidate references with source labels; never silently claim a found design is the user's exact original. Generated designs should be labeled 'Keepsake recreation' or 'commemorative stub' in metadata and should omit scannable barcodes/QR codes and sensitive pass identifiers.
Core data	ticket_artifact, event_name, venue, event_date, seat_text, source_pass_id, reference_sources, reconstruction_type, provenance_note, generated_asset.
Guardrails	Exact historical ticket matching may be impossible because many ticket

	designs are personalized or unavailable publicly. The system should distinguish authentic reference, approximate reference, and generated recreation. Generated stubs are memory artifacts, not valid admission credentials.
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OPEN DECISION / NEEDS MORE ATTENTION: For Apple Wallet integration, determine whether launch uses user-shared pass files/screenshots, manual metadata entry, or a native-device integration. Browser access to Wallet data should not be assumed.

30.5 Scrapbook Artifacts

Purpose	Allow Memory Box objects to become page elements alongside photos, captions, and stickers.
Experience	An artifact can look like a ticket tucked under a photo, a folded letter, a newspaper clipping, a child's drawing, or a postcard. Front/back objects can be flipped.
Implementation direction	Unify movable page content under a PageElement model with types such as photo, sticker, caption, and artifact. Reuse transform, z-order, selection, and undo/redo infrastructure.
Core data	page_element.type, artifact_id, x, y, rotation, scale, z_index, displayed_side.
Guardrails	Artifacts should inherit the same privacy rules as the scrapbook/page in which they appear unless explicitly restricted further.

30.6 Memory Trails

Purpose	Create gentle paths through related memories without resembling recommendation feeds.
Experience	A small handwritten cue such as 'This reminds me of something...' can connect the current memory to another book, page, person-tag, place, date, tradition, or manually linked memory.
Implementation direction	Start with manual links and deterministic metadata relationships. Later, optional semantic suggestions can propose connections. A trail opens one destination at a time rather than an endless recommendation list.
Core data	memory_links, source_id, destination_id, relation_type, user_confirmed, display_text.
Guardrails	No engagement ranking, popularity optimization, infinite recommendations, or automatic public exposure.

30.7 Desk Drawer Time Capsules

Purpose	Let users intentionally send memories to their future selves.
Experience	A user places photos, artifacts, a note, or a scrapbook page into an envelope and chooses an unlock date. When the date arrives, an envelope quietly appears on the desk.
Implementation direction	Store an encrypted/permission-protected capsule manifest and unlock_at timestamp. The contents remain inaccessible through normal application routes until the unlock condition is satisfied. Presentation is environmental, not a red-badge notification.
Core data	time_capsules, owner_id, unlock_at, contents, created_at, opened_at, envelope_style.
Guardrails	Account recovery/export policy needs special consideration: a user's own data export should not accidentally undermine the intended lock without clearly explaining the tradeoff.

30.8 Seasonal Room Memory

Purpose	Allow the room to develop traditions by remembering selected seasonal choices.
Experience	If the user places a small ornament or seasonal object one year, Keepsake may quietly return that object the next time the season arrives. The room feels like it remembers traditions.
Implementation direction	Track only eligible seasonal decor and explicit/learned placement preferences. Reapply a small number of items at season transitions. Give the user an easy 'don't remember this' control.
Core data	seasonal_memories, decor_asset_id, season, placement_slot, last_used, auto_return_enabled.
Guardrails	Do not restore clutter indiscriminately. Curated RNG and the room's object budget still apply.

30.9 Lights Out Mode

Purpose	Provide a distraction-free reading mode that exists naturally inside the room.
Experience	Clicking the desk lamp can darken the surrounding room, hide nonessential interface chrome, and leave the scrapbook illuminated by the lamp, CRT glow, and/or window.
Implementation direction	Implement as a room/view state that reduces UI, changes lighting presets, and suppresses ambient events. The scrapbook remains fully

	operable. Provide a clear keyboard/mobile/accessibility equivalent.
Core data	view_mode, lights_out_enabled, preferred_lamp_level, ui_visibility.
Guardrails	Never make essential navigation invisible. Reduced-vision and high-contrast settings override aesthetic darkness when necessary.

30.10 Photograph Backs

Purpose	Replace conventional metadata sidebars with a tactile place for personal context.
Experience	A photograph can be flipped over to reveal handwriting-style notes, date, place, tagged people, a longer memory, or archival information. Users decide what is visible to visitors.
Implementation direction	Store photo-back content independently from file metadata. The flip is a presentation state; in reduced-motion mode it becomes an immediate front/back switch. Support both user-authored text and selected imported metadata.
Core data	photo_back, photo_id, note, date_label, place_label, people_tags, visibility, handwriting_style.
Guardrails	EXIF data, especially exact GPS, must never be exposed to visitors merely because it exists in the original file.

30.11 Found Photos

Purpose	Gently resurface uploads that the user intended to use but never actually touched.
Experience	Keepsake may occasionally place a small 'found photographs' envelope in the filing area containing a few uploads that have remained untouched since import. There is no streak, badge, reward, or urgency.
Implementation direction	Track meaningful interaction rather than merely upload age. A photo becomes eligible only after a long quiet period if it has never been placed in a scrapbook, moved/edited, captioned, favorited, categorized, or otherwise intentionally used. Apply long cooldowns and allow the user to disable rediscovery.
Core data	photo_activity, uploaded_at, first_used_at, last_meaningful_action, found_photo_eligible, last_resurfaced_at.
Guardrails	Do not resurface deleted/hidden/private-sensitive items, recently uploaded items, or photos the user has explicitly dismissed from rediscovery.

30.12 Room Sound Geography

Purpose	Make the future room feel physically coherent through positional and zone-based sound.
Experience	Near the window, rain/wind/birds become slightly clearer; near the CRT, music has stronger presence; at the desk, page turns and paper sounds feel close; the filing cabinet has subtle drawer/folder sounds.
Implementation direction	Use a small number of positional or zone-based audio sources rather than a complex simulation. Crossfade smoothly as the camera changes authored viewpoints. Keep ambience, music, UI, and interaction volumes independently adjustable.
Core data	audio_zones, source_position, category, gain, falloff, user_volume_preferences.
Guardrails	Sound must never carry essential information. Respect mute, reduced sensory settings, and mobile battery/performance constraints.

30.13 How These Systems Fit Together

- Memory Box is the archival home for non-photo keepsakes; Scrapbook Artifacts are their page-level representations.
- Photograph Backs and artifact backs preserve the tactile idea that objects can contain information on both sides.
- Memory Trails connect photos, artifacts, pages, books, places, dates, and traditions without creating a feed.
- Found Photos works as a low-pressure archive rediscovery mechanism specifically for untouched uploads.
- Time Capsules deliberately hide selected memories until a future date; Found Photos does the opposite by resurfacing forgotten material.
- Memory Lighting, Seasonal Room Memory, Lights Out Mode, and Room Sound Geography deepen the existing room instead of adding more furniture.
- The AI Ticket Companion belongs inside the Memory Box workflow and should preserve a clear distinction between original source material, web references, and generated commemorative reconstructions.

30.14 Data Model Consequence

Important architecture change: The long-term page editor should not be built around 'photos on a page.' It should be built around generic PageElements. Photos, artifacts, stickers, captions, and future keepsakes can then share selection, movement, rotation, scale, layering, visibility, undo/redo, and persistence.

PageElement
 - id
 - pageId

- type: photo | artifact | sticker | caption | keepsake
- sourceId
- x / y
- width / height / scale
- rotation
- zIndex
- frontBackState
- visibility
- locked
- createdAt
- updatedAt

30.15 Updated Build Priority

- NOW: Finish refactor and multi-photo PageElement foundation.
- NEXT: Captions, multiple photos, z-order, undo/redo, local persistence, better page turns.
- EARLY EXPANSION: Photograph Backs and generic Artifact support, because both affect the core data model.
- AFTER PERSISTENT BACKEND: Memory Box archive, Found Photos, Memory Trails, and Time Capsules.
- AFTER ROOM SHELL: Lights Out, Memory Lighting, Seasonal Room Memory, and Room Sound Geography.
- LATER / RESEARCH-HEAVY: AI Ticket Companion, automated memory suggestions, sophisticated contextual ambience.

30.16 Updated Inspiration Note

- LittleBigPlanet - tactile craft, playful physical materials, handmade personality.
- Sly Cooper and the Thievius Raccoonus - stylized visual identity and readable, authored spaces.
- Life is Strange - intimate memories, photography, youth/nostalgia, environmental mood.
- What Remains of Edith Finch - environmental storytelling, restrained diegetic text, memories embedded in spaces.
- Firewatch - warm stylized lighting, quiet exploration, strong sense of place, ambience, color, and environmental storytelling.